



Data Engineering **Hackathon** 2024

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Team Name: base234

Team Size: 4

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Bhattacharyya, Debankita Dutta

Problem Statement: AI/ML and Data Science

Problem Statement Theme: Healthcare Diagnosis

Idea Brief

Hitaya is a one-stop solution, a medical diagnosis App that can allow users to upload medical data or x-ray scans, and the APP will predict whether a patient has any disease or not.

Machine Learning has been revolutionizing on healthcare domain. ML models can now detect patterns underlying diseases. In this way, AI techniques can be considered as the second pair of eyes for doctors, that can decode patient health knowledge extracted from large data sets by summing up facts & observations of diseases.

Features

- Creating an AI/ML preliminary diagnosis application.
- Users/Doctors can input images of x-ray scans/ medical data/ or just images and our Application will give them the diagnosis results.
- Using Generative AI to give user health Recommendations or Provide Diet Plans.
- Diagnostic System to predict the risk of a disease based on user data.
- Using Informatica create an ML pipeline that will re-train ML models from new data and upload the models using Model Serve for our UI to consume.

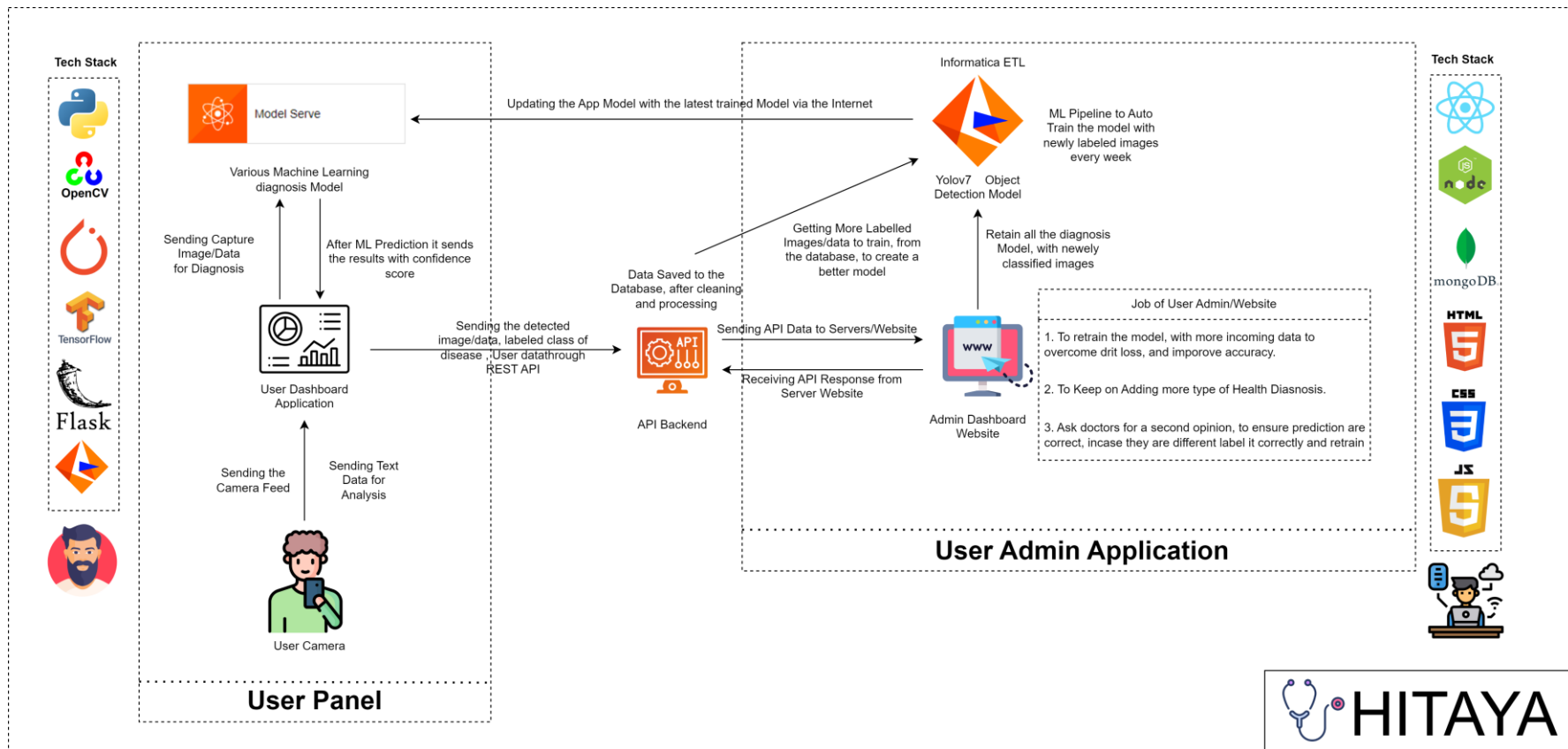
Health Analysis

 Breast Cancer Prediction Use ML to Predict the chances of Breast Cancer. Let's check	 Chronic Kidney Disease Use ML to Predict the chances of Kidney Disease. Let's check	 Diabetic Prediction Use ML to Predict the chances of Diabetes. Let's check	 Heart Disease Use ML to Predict the chances of Heart Disease. Let's check
 Liver Disease Use ML to Predict the chances of Liver Disease. Let's check	 Chest X-Ray Scan Use ML to Predict the chances of disease from Chest-Xray Scans. Let's check	 Skin Cancer Use ML to Predict the chances of skin Cancer. Let's check	 Detect Pneumonia X-Ray Use ML to Predict the chances of Pneumonia from Chest-Xray. Let's check

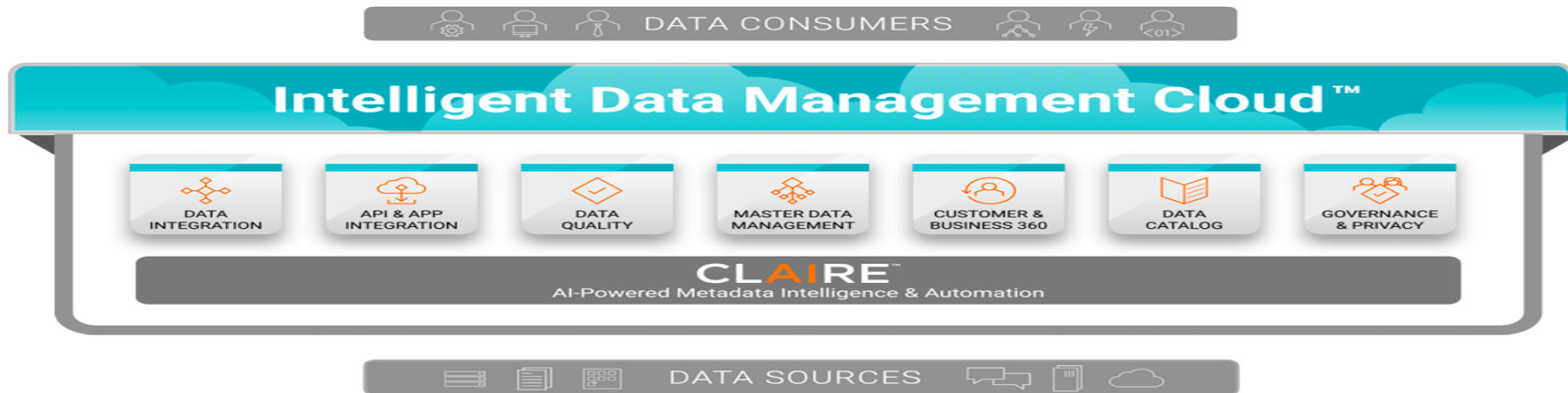
Tech Stack

1. Flask
2. React
3. Informatica
4. Snowflake
5. Machine Learning (Pytorch, scikit-learn, Tensorflow)
6. Node.js

Project Architecture

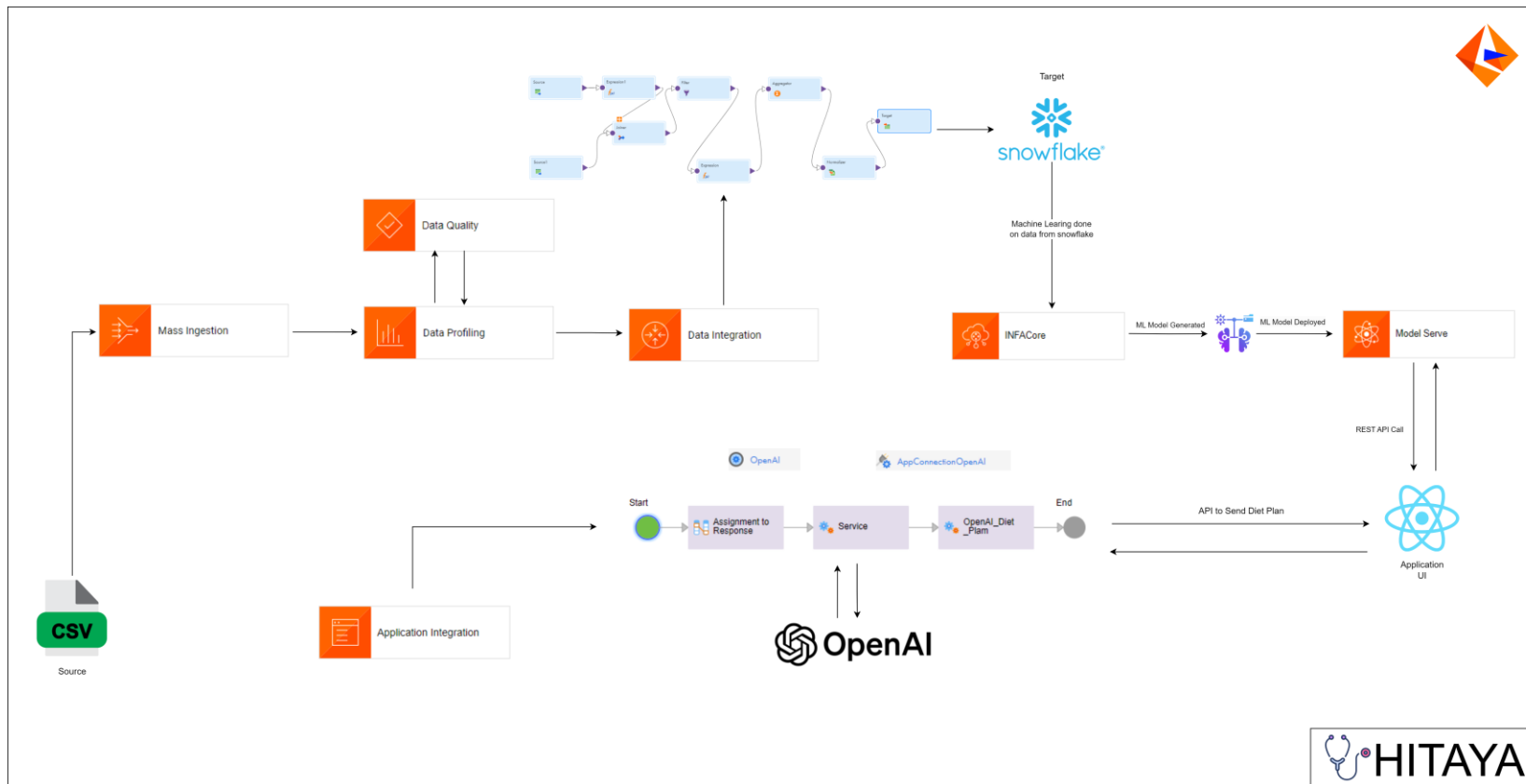


What positive and unique solutions your Idea have?



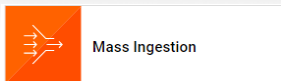
With Informatica's Intelligent Data Management Cloud, all these services are available to natively on a cloud-first, microservices-based, API-driven platform that's both elastic and serverless. CLAIRE, AI technology, is embedded in every service. Data can be brought from any source, of any type, to serve any user anywhere.

Project Architecture Informatica



Creating a machine learning (ML) pipeline for training disease diagnosis data from Snowflake using Informatica INFACore involves several steps. Here's a high-level overview of the process:

- 1. Data Collection:** Open Source and readily available data is collected from various sources. This primarily contains patients historical data on large scale of factors influencing a particular disease. The data is saved into CSV as **Flat files and ingested to Informatica Mass Ingestion**.
- 2. Data Processing:** Clean and preprocess the data using Informatica's, **Data Profiling, Data Quality** checks. This may involve **normalizing data, handling missing values, and creating features** that will be used for training the ML model.
- 3. Data Loading:** Load the **processed data into Snowflake tables using Data Integration**. Ensure that the data is stored in an organized manner that's conducive to analysis and model training.
- 4. Model Training:** Utilize Informatica **INFACore** to **read, write and use frequently available functions over pandas dataframes** for model training. Made use of popular ML frameworks like scikit-learn, XGBoost, and LightGBM for training.
- 5. Model Deployment:** After training, **deploy your model using ModelServe**. This includes managing models with **Model Registry**, which allows secure deployment and management of models in Informatica.
- 6. Model Inference:** **Leverage the deployed model through ModelServe Deployment RestAPI**. This can be done by setting up a pipeline that automatically feeds new data to the model and retrieves predictions.
- 7. Monitoring and Management:** Continuously **monitor the performance of your ML model and manage it privileges**. This includes updating the model with new data and retraining if necessary.



t_heart_disease_data_ingestion

1 Definition 2 Source 3 Target 4 Actions 5 Runtime Options

Source Details

Source Type:* ☒ Source Connection ☐ File Listener

Connection Type:*

Source Options

File Pickup:* ☒ By Pattern ☐ By File List

Source Directory:* [Add Parameters](#)

☐ Include files from sub-folders

☒ File Pattern	Wildcard	*.heart.csv
☐ File Date		
	Timezone:	
☐ File Size		

☐ Skip Duplicate Files

☐ Check File Stability Stability Check Interval:

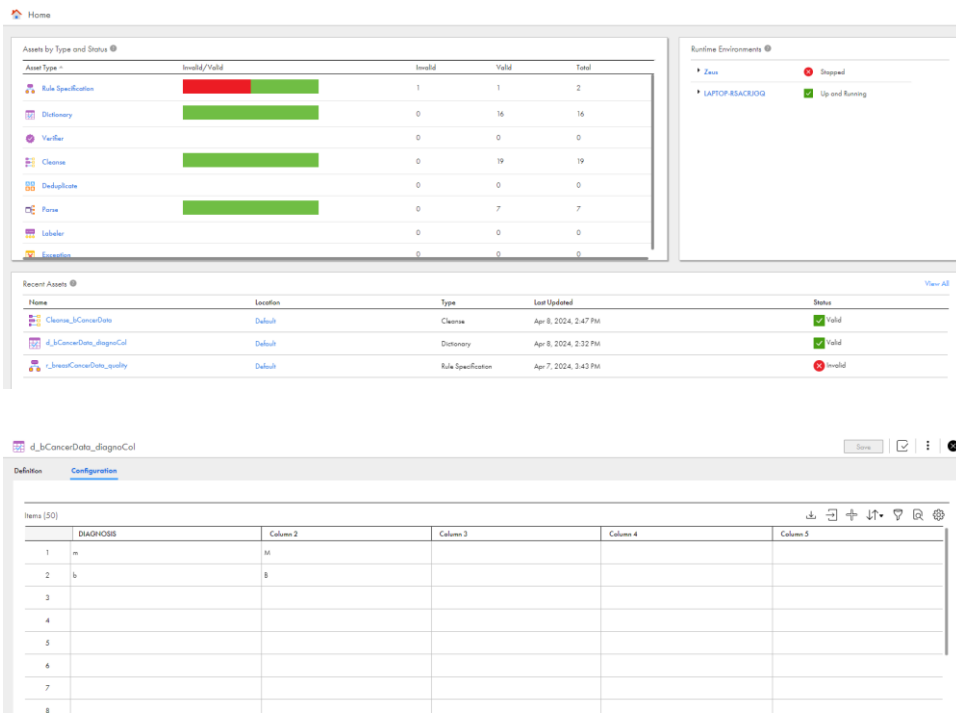
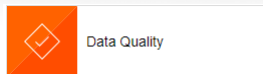
Batch Size:*

After File Pickup:

Mass ingestion involves the systematic transfer and replication of substantial volumes of data, typically between different databases or repositories, to facilitate their utilisation or storage. To efficiently manage the ingestion or replication of extensive data sets between a relational database and a Hive or HDFS target.

1. Here are the few bunch of csv data file for different health disease predictions which we load into the respective snowflake tables using flat file mass ingestion.
2. Here's we configure the source connection pointing to the local directory of our target dataset
3. Here's the target connection configuration which indicates the destination where the dataset will be finally loaded through mass ingestion and a snowflake object is being configured as the target destination

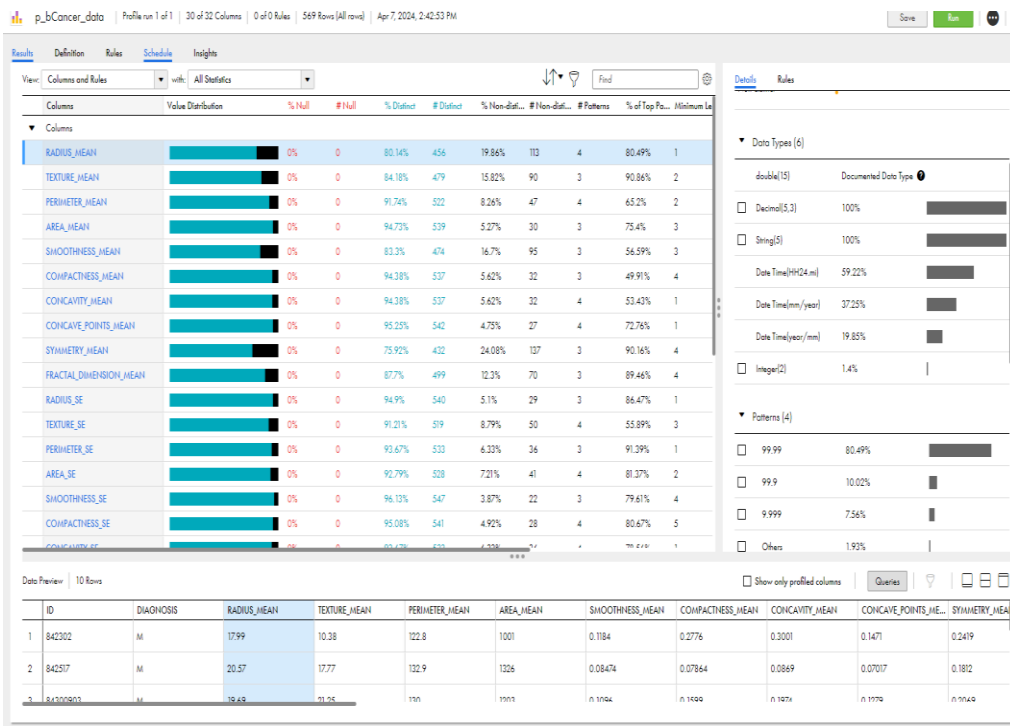
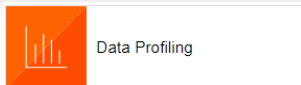
That's how Informatica Mass Ingestion helps us to seamlessly integrate with the broader Informatica Intelligent Data Platform, allowing us to leverage additional capabilities such as data integration, data quality, metadata management, and master data management to enhance our data management processes once the data is reached to the configured target destination



Data quality we maintain the accuracy, completeness, consistency, and reliability of data stored in sources and used for training AI Modls.

- ✓ Here the data that we got from the csv dataset of breast cancer predictions ,we applied few cleansing mechanism in order to enhance the quality of the desired data
- ✓ We have applied one of the predefined rules which is removing spaces from our raw dataset, so that the string type values doesn't added with any leading or trailing spaces
- ✓ Here's a sample test_data where we performed the above rule on to check if it produces the desired result
- ✓ The below output shows how it cleanse the input data and produces the expected result
- ✓ Here we have standardized our data further using the dictionary where we specified the valid values of diagnosis column to M and B and whenever this column receives the data other than that it will simply standardize the data to the value set to the output column

That's how data quality in Informatica encompasses a range of activities and capabilities aimed at improving the accuracy, completeness, and reliability of our raw_data set for different diseases.



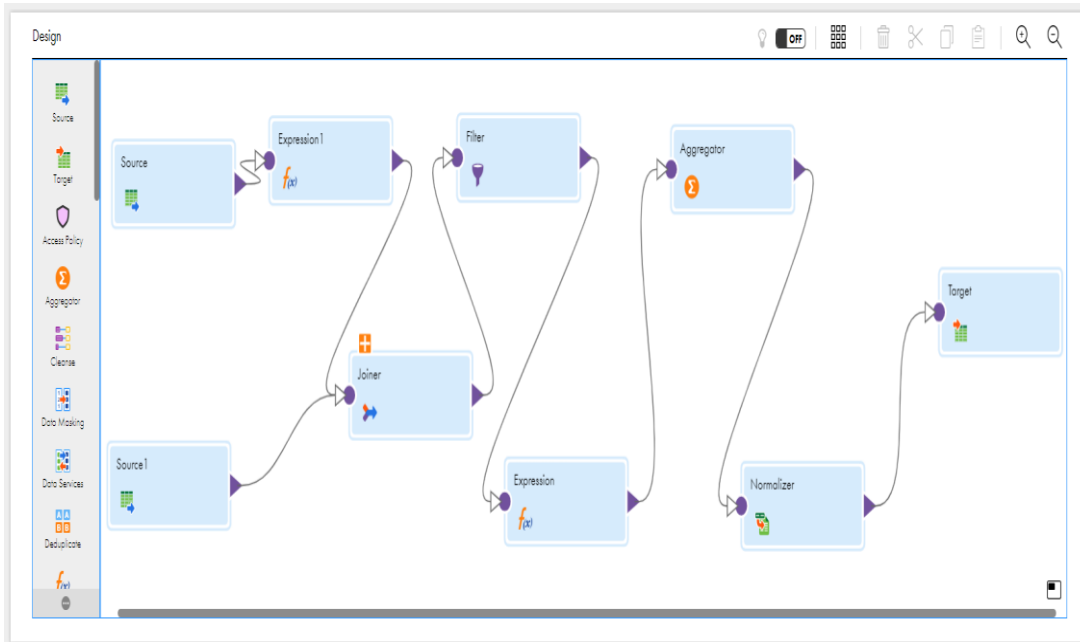
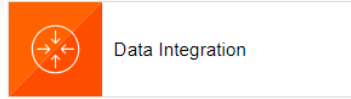
- Data Profiling to assess the quality of data based on our requirements.
- Analysing data to understand it's content, structure and patterns.
- Identify anomalies, inconsistencies, and potential issues in the data

✓ Content Assessment:

- Data profiling examined value frequencies, data types, and distribution.
- It helped us to understand what data exists, how often it occurs, and whether it aligns with our expectations.

✓ Structure Assessment:

- Data profiling identifies keys, functional dependencies, and relationships between data elements.
- It ensures that data adheres to expected structures (e.g., primary keys, foreign keys).

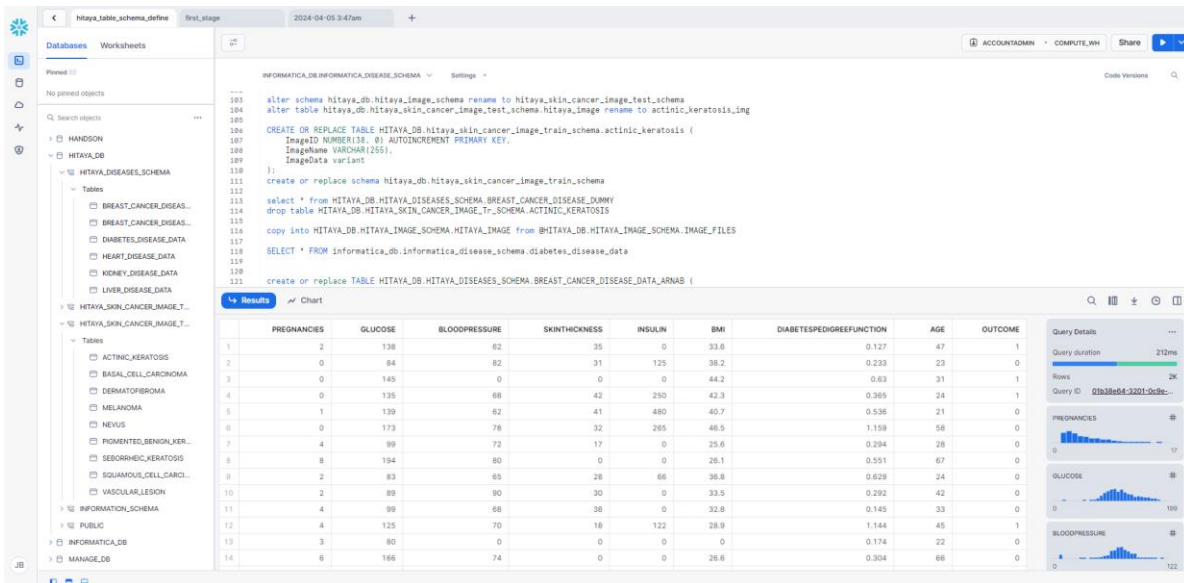


At Data Integration we have combined Data from Different Sources and streamlined it to upload it to Snowflake Data Warehouse.

- **Joiner:** joined data form different sources.
- **Filter:** Filter data from the sequence flow.
- **Expressions:** Performed calculation on individual rows of data.
- **Aggregator:** Performed aggregate on our data.
- **Normalizer:** normalised the data.



IDMC provides a Snowflake cloud Connector, which allows us to integrate Snowflake with IDMC workflows. We used this connector to extract, transform, and load (ETL) data IDMC to load source data from databases, applications, and data files in the cloud or on-premises into Snowflake.



The screenshot shows the Snowflake SQL Editor interface. The top bar indicates the current database is 'INFORMATICA_DB' and the schema is 'INFORMATICA_DISEASE_SCHEMA'. The query being executed is as follows:

```

103 alter schema hitaya_db.hitaya_image_schema rename to hitaya_skin_cancer_image_test_schema
104 alter table hitaya_db.hitaya_skin_cancer_image_test_schema.hitaya_image rename to actinic_keratosis_img
105
106 CREATE OR REPLACE TABLE HITAYA_DB.hitaya_skin_cancer_image_train_schema.actinic_keratosis (
107   ImageID NUMBER(10, 0) AUTOINCREMENT PRIMARY KEY,
108   ImageName VARCHAR(255),
109   ImageData variant
110 );
111
112 create or replace schema hitaya_db.hitaya_skin_cancer_image_train_schema
113
114 select * from HITAYA_DB.HITAYA_DISEASES_SCHEMA.BREAST_CANCER_DISEASE_DUMMY
115 drop table HITAYA_DB.HITAYA_SKIN_CANCER_IMAGE_Tr_SCHEMA.ACTINIC_KERATOSIS
116
117 copy into HITAYA_DB.HITAYA_IMAGE_SCHEMA.HITAYA_IMAGE from @HITAYA_DB.HITAYA_IMAGE_SCHEMA.IMAGE_FILES
118
119 SELECT * FROM Informatica_DB.Informatica_disease_schema.diabetes_disease_data
120
121 create or replace TABLE HITAYA_DB.HITAYA_DISEASES_SCHEMA.BREAST_CANCER_DISEASE_DATA_ARNAB (

```

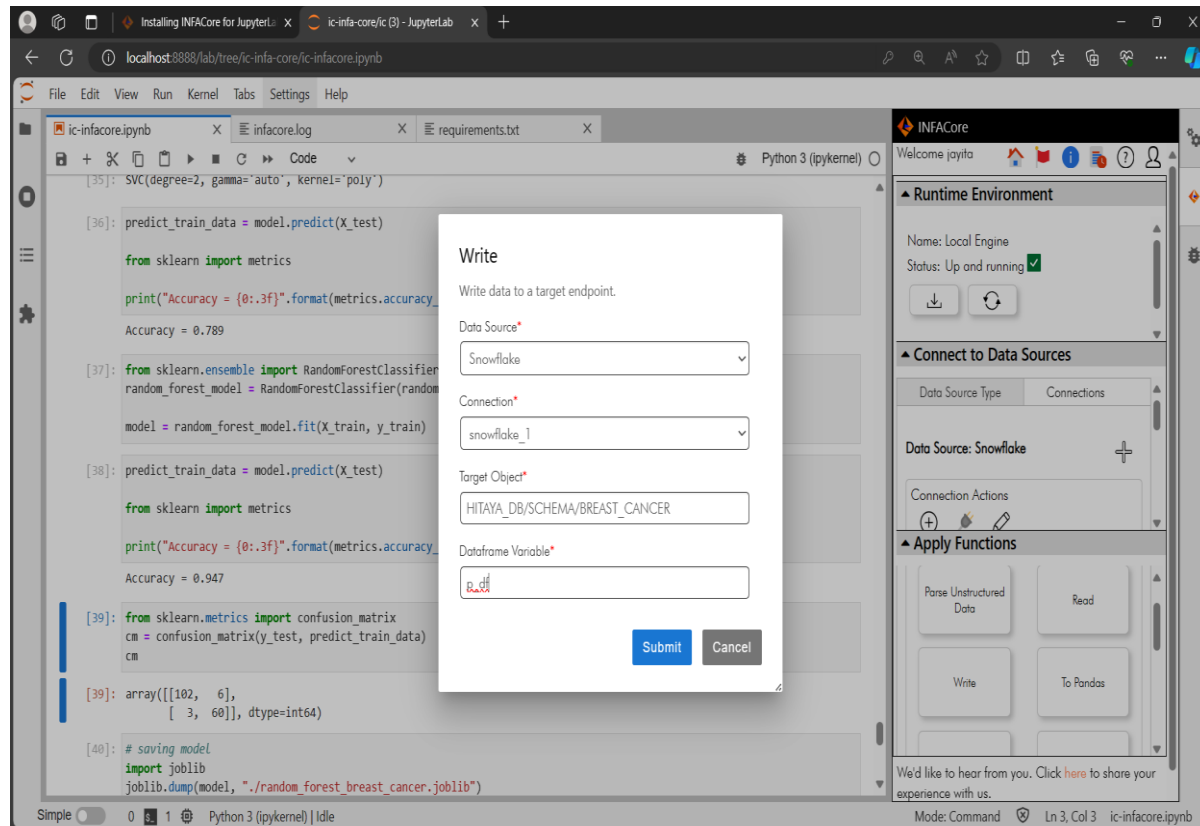
The results of the query are displayed in a table with 14 columns: PREGNANCIES, GLUCOSE, BLOODPRESSURE, SKINTHICKNESS, INSULIN, BMI, DIABETESPEDIGREEFUNCTION, AGE, and OUTCOME. The table contains 14 rows of data.

	PREGNANCIES	GLUCOSE	BLOODPRESSURE	SKINTHICKNESS	INSULIN	BMI	DIABETESPEDIGREEFUNCTION	AGE	OUTCOME
1	2	138	82	35	0	33.6	0.127	47	1
2	0	84	82	31	125	38.2	0.233	23	0
3	0	145	0	0	0	44.2	0.63	31	1
4	0	135	68	42	250	42.3	0.365	24	1
5	1	139	62	41	480	40.7	0.536	21	0
6	0	173	78	32	265	46.5	1.159	58	0
7	4	99	72	17	0	25.6	0.294	28	0
8	8	194	80	0	0	26.1	0.551	67	0
9	2	83	65	28	66	36.8	0.629	24	0
10	2	89	80	30	0	33.5	0.292	42	0
11	4	99	68	0	32.8	0	0.145	33	0
12	4	125	70	18	122	38.9	1.144	45	1
13	3	80	0	0	0	0	0.174	22	0
14	6	166	74	0	0	26.6	0.304	66	0

- ✓ First of all we configured a snowflake connection to our IDMC data platform by specifying the account Name, credentials, warehouse and runtime environment .
- ✓ Then we created a mapping to read comma delimited csv files from local, and write the results into the respective tables in Snowflake. This dataset for diabetes predictions is now loaded to its respective table in snowflake.



INFACore



The screenshot shows a JupyterLab environment with a Python 3 (ipykernel) session. The code in the notebook performs the following steps:

- Imports `SVC` from `sklearn.svm` and `metrics` from `sklearn`.
- Trains a model using `SVC(degree=2, gamma='auto', kernel='poly')` on `X_train` and `y_train`.
- Predicts on `X_test` and prints the accuracy: `print("Accuracy = {:.3f}".format(metrics.accuracy_score(y_test, predict_train_data)))`. The output is `Accuracy = 0.789`.
- Imports `RandomForestClassifier` from `sklearn.ensemble` and trains a model using `random_forest_model = RandomForestClassifier(random_state=42)` on `X_train` and `y_train`.
- Predicts on `X_test` and prints the accuracy: `print("Accuracy = {:.3f}".format(metrics.accuracy_score(y_test, predict_train_data)))`. The output is `Accuracy = 0.947`.
- Imports `confusion_matrix` from `sklearn.metrics` and calculates the confusion matrix: `cm = confusion_matrix(y_test, predict_train_data)`.
- Prints the confusion matrix: `print(cm)`. The output is `array([[102, 6], [ 3, 60]], dtype=int64)`.
- Saves the model using `joblib.dump(model, './random_forest_breast_cancer.joblib')`.

A 'Write' dialog box is open, prompting the user to write data to a target endpoint. The fields are filled as follows:

- Data Source: `snowflake`
- Connection: `snowflake_1`
- Target Object: `HITAYA_DB/SCHEMA/BREAST_CANCER`
- Dataframe Variable: `model`

The dialog box has 'Submit' and 'Cancel' buttons.

- **INFACore** to access data from diverse data sources, explore data, apply functions, transform the data, and deliver quality data to your business.
- **INFACore SDK** for Python.
- **JupyterLab** extension for **INFACore**.
- Leveraging above tools to build machine learning models for different kinds of disease predictions by injecting data from flat files and saving transformations/predictions to Snowflake database.



Model Serve



Deployment_Breast_Cancer

Stop Deployment

Restart Deployment



Overview

Test

Overview

Status	Location	Requests	Duration	Description	Max compute ...	Created on	Start time	Last updated	Started by	Updated by	Owner
Available	Default	4	4 hours 17 m...		4	Apr 6, 2024, ...	Apr 6, 2024, ...	Apr 7, 2024, ...	manijb13@g...	manijb13@g...	manijb13@g...

Model endpoint URL

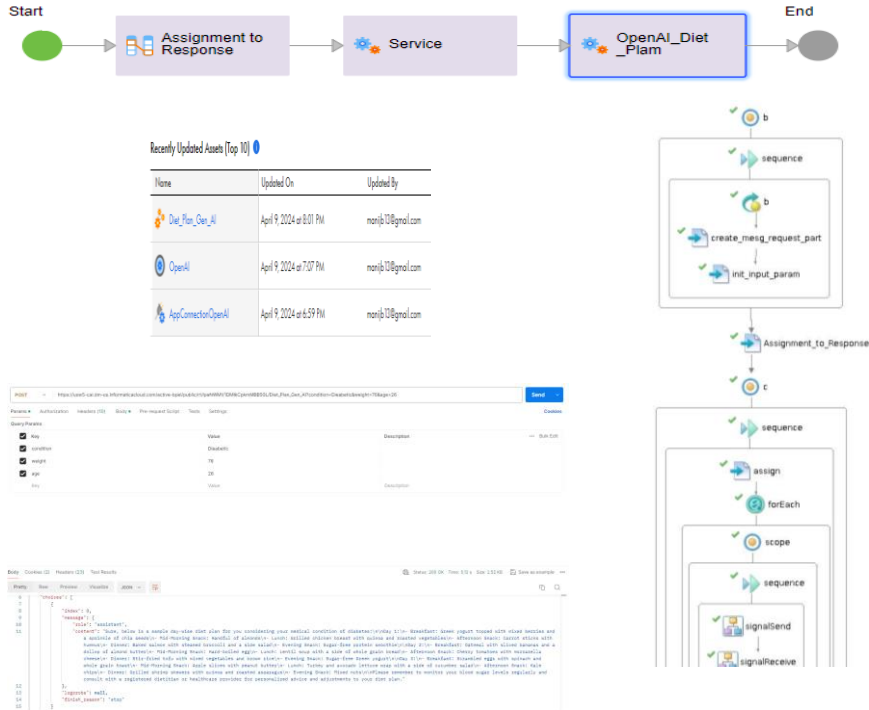
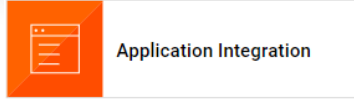
REST API endpoint to communicate with the machine learning model.

<https://usw5-dsml.dm-us.informaticacloud.com/ml-predict/api/v1/deployment/8AkKMrFL13FhZ3n47KmswN>

Machine learning model

Breast-Cancer

- Simplifying MLOps with Informatica ModelServe.
- Model Serve to deploy machine learning models and generate predictions based on historical data.
- Monitor, alert and consume served AI/ML models in applications.



React App is calling Application Integration published API, to get the diet plans.

- **Service Connector using Form:** Connect to OpenAI Rest endpoint, sending age, weight and health condition and prompt template as to fetch the ChatGPT response with a Day wise Diet plan.
- **App Connection:** perform actions to connect our process to a service connector .

ML Diagnosis System

Hitaya is a one-stop solution, a medical diagnosis App that can allow users to upload medical data or x-ray scans, and the APP will predict whether a patient has any disease or not.

List of Diseases it can predict now

- Breast Cancer
- Chronic Kidney Disease
- Diabetics
- Heart Disease
- Liver Disease
- Chest X-ray Scans
- Skin Cancer
- Pneumonia X-Ray

Breast Cancer

Breast cancer is a disease in which cells in the breast grow out of control.

There are different kinds of breast cancer. The kind of breast cancer depends on which cells in the breast turn into cancer.

Breast cancer can begin in different parts of the breast.

There are certain medical parameters which will define whether the female individual is affected with breast cancer.

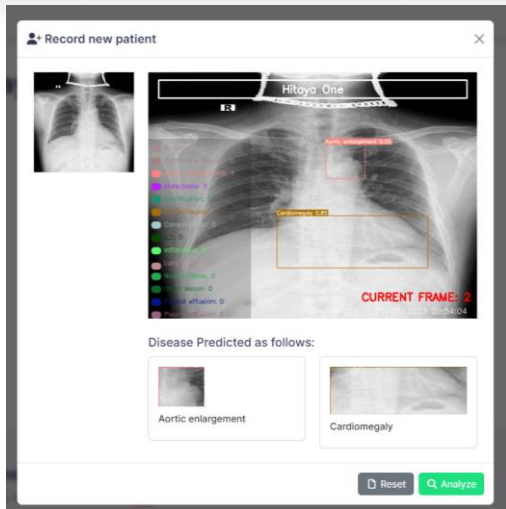
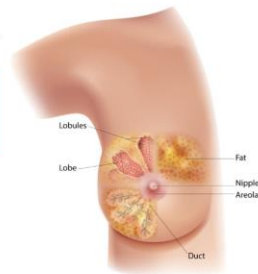
[Click here to predict breast cancer.](#)

A breast is made up of three main parts:

- **Lobules:** glands that produce milk.
- **Ducts:** tubes that carry milk to the nipple.
- **Connective tissue:** consists of fibrous and fatty tissue that surrounds and holds everything together.

Most breast cancers begin in the ducts or lobules. It can spread outside the breast through blood vessels and lymph vessels. When it spreads to other parts of the body, it is said to have **metastasized**.

Radius Mean





Skin cancer type

Melanoma

Description:

Treatment options for melanoma depend on the stage and location of the cancer but may include:

- Surgical excision:** The primary treatment for early-stage melanoma involves surgically removing the tumor and a margin of healthy skin around it.
- Lymph node biopsy:** If melanoma has spread or is at a high risk of spreading, nearby lymph nodes may be tested for the presence of cancer cells.

Record new patient

Full name:

Age: Gender:

Breast Data

Radius Mean: 17.99	Texture Mean: 10.38	Perimeter Mean: 122.8
Area Mean: 1001	Smoothness Mean: 0.1184	Compactness Mean: 0.2776
Concavity Mean: 0.3001	Concave Points Mean: 0.1471	Symmetry Mean: 0.2419
Fractal Dimension Mean: 0.07871		
Radius SE: 1.095	Texture SE: 0.9053	Perimeter SE: 8.589
Area SE:	Smoothness SE:	Compactness SE:

Diet Plans

Use of **GenAI** to generate health plans, diet tips, exercise routine customized according to user input received, as per user's health report.

Weight:

67

Age:

54

Condition:

Diabetic

Reset Form

Sample Fill

Generate

Recommended Diet Plan for an Individual of age 54, weighs around 67 having a Diabetic condition as follows:

Maintain a healthy weight by eating a balanced diet and exercising regularly. Limit your intake of unhealthy fats, sugar, and processed foods. Choose foods that are high in fiber, protein, and vitamins. Exercise for at least 30 minutes most days of the week. Talk to your doctor about the best type of exercise for you.

Resources



<https://www.youtube.com/watch?v=H3Yu-IWlo1E>



<https://github.com/IntelegixLabs/Informatica>



Data Engineering
Hackathon
2024

THANK YOU

